

C.4.4 Incentives

This section tests if incentives for assessment accuracy promote radiologists to make more accurate assessments. We find that the incentives do not play a significant role in getting a correct response. The effect of incentives are estimated using the following regression specification and the results are shown in Table C.14.

$$\begin{aligned}
Y_{irt} = & \gamma_{h_i} + \gamma_{INC} \cdot d_{INC}(r) \\
& + \gamma_{CH} \cdot d_{CH}(t) + \gamma_{CH \times INC} \cdot d_{CH}(t) \cdot d_{INC}(r) \\
& + \gamma_{AI} \cdot d_{AI}(t) + \gamma_{AI \times INC} \cdot d_{AI}(t) \cdot d_{INC}(r) \\
& + \gamma_{AI \times CH} \cdot d_{CH}(t) \cdot d_{AI}(t) + \gamma_{AI \times CH \times INC} \cdot d_{CH}(t) \cdot d_{AI}(t) \cdot d_{INC}(r) + \varepsilon_{irt}
\end{aligned}$$

where Y_{irt} is an outcome variable of interest for radiologist r diagnosing patient case-pathology i and treatment t , and γ_{h_i} are pathology fixed effects. Here CH refers to cases with access to clinical history information, AI to cases with AI predictions and INC refers to incentivized cases.

Table C.14: Effect of Incentives

	Deviation from AI		Deviation from GT		Effort Measures	
	Top-Level with AI	Pooled with AI	Top-Level with AI	Pooled with AI	Active Time	Clicks
Treatment	(1)	(2)	(3)	(4)	(5)	(6)
AI $\times$ CH	−0.002 (0.007)	−0.003 (0.003)	0.002 (0.013)	0.001 (0.004)	−8.71 (8.30)	−1.18 (2.28)
AI	−0.032 (0.007)	−0.013 (0.003)	0.005 (0.009)	0.001 (0.004)	15.46 (6.51)	3.57 (1.66)
CH	0.001 (0.005)	0.001 (0.002)	−0.017 (0.009)	−0.007 (0.004)	10.90 (5.24)	1.01 (1.59)
CONTROL MEAN	0.225 (0.008)	0.113 (0.004)	0.219 (0.012)	0.083 (0.006)	159.47 (9.63)	39.93 (2.07)
AI $\times$ CH $\times$ INCENTIVES	0.009 (0.010)	0.006 (0.004)	0.006 (0.017)	−0.001 (0.006)	17.51 (13.79)	3.00 (3.26)
AI $\times$ INCENTIVES	−0.017 (0.009)	−0.007 (0.004)	−0.009 (0.012)	0.000 (0.005)	−18.42 (9.25)	−4.09 (2.07)
CH $\times$ INCENTIVES	−0.008 (0.008)	−0.004 (0.004)	0.007 (0.014)	0.005 (0.005)	−8.75 (9.54)	−1.83 (2.16)
CONTROL MEAN $\times$ INCENTIVES	0.005 (0.010)	0.002 (0.004)	0.004 (0.013)	−0.002 (0.005)	4.17 (14.09)	1.56 (3.05)
PATHOLOGY FE	Yes	Yes	Yes	Yes	-	-
OBSERVATIONS	20440	132860	20440	132860	6718	6718
F-STAT	.95	.94	.49	.41	1.09	1.16
P>F-STAT	.44	.44	.74	.80	.36	.33

Note: This table summarizes the average treatment effects (ATE) of different information environments on the (1) absolute value of the difference between the radiologist probability and AI algorithm probability (Columns (1) and (2)), absolute value of the difference between the radiologist probability and the ground truth (Columns (3) and (4)) and radiologists' effort measured in terms of active time and clicks for all and the second-half images (Columns (5) and (6)). The F-statistic tests for the joint significance of the four incentivized groups. Top-level specification includes two pathologies: airspace opacity and cardiomedastinal abnormality while Pooled AI includes all the pathologies with AI predictions excluding abnormality and support device hardware. Only cases in design 1 and design 3 are considered. Two-way clustered standard errors at the radiologist and patient-case level are in parenthesis.

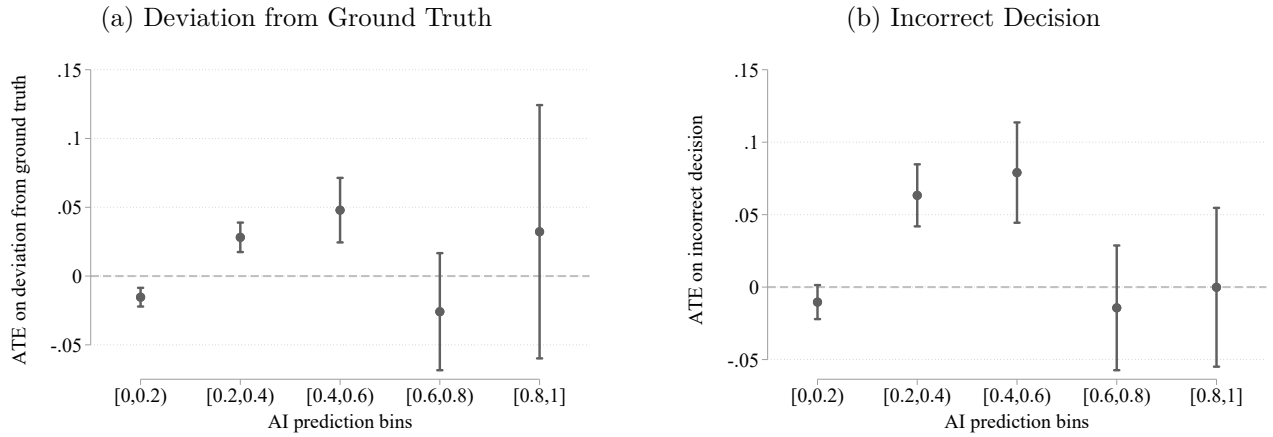
C.4.5 Controlling for sequence number and session

Figure C.29 uses the following specification that controls for the sequence number in which the participants saw a particular case within one experiment session and the session dummies for the different designs and experiment sessions to estimate the heterogeneous treatment effects. There are four sessions in Design 2, whereas Design 1 and 3 have only one session.

$$Y_{irt} = \gamma_{h_i} + \gamma_{AI} \cdot d_{AI}(t) + \sum_g \left[\gamma_g \cdot d_g(s_i^A) + \gamma_{AI \times g} \cdot d_{AI}(t) \cdot d_g(s_i^A) \right] + \gamma_{w_{irt}} + \gamma_{m_{irt}} + \varepsilon_{irt}$$

where Y_{irt} is an outcome variable of interest for radiologist r diagnosing patient case-pathology i and treatment t , γ_{h_i} are pathology fixed effects, $\gamma_{w_{irt}}$ are sequence number dummies and $\gamma_{m_{irt}}$ are session dummies. Here, g is defined as an index for an interval of the AI signal range where $1 \leq g \leq 5$. A case is said to be in an interval g conditional on the signal value for the given patient-case.

Figure C.29: Conditional treatment effect given AI prediction



Note: Main specifications similar to figure 3 with additional controls for rounds and session.

Table C.15: Average Treatment Effects

Sessions	Deviation from AI	Deviation from GT	Effort Measures	
	(1)	(2)	Active Time	Clicks
	(3)	(4)		
DESIGN 2: SESSION 1	−0.016 (0.011)	0.029 (0.014)	41.18 (11.17)	14.72 (2.68)
DESIGN 2: SESSION 2	−0.033 (0.010)	0.014 (0.011)	−3.68 (9.91)	6.64 (2.48)
DESIGN 2: SESSION 3	−0.032 (0.011)	0.004 (0.012)	−25.13 (9.77)	3.68 (2.58)
DESIGN 2: SESSION 4	−0.035 (0.010)	0.007 (0.011)	−38.73 (7.87)	1.35 (2.38)
DESIGN 3	0.020 (0.009)	−0.005 (0.012)	- -	- -
CONTROL MEAN	0.219 (0.007)	0.216 (0.010)	164.82 (6.74)	40.76 (1.53)
DESIGN 2: SESSION 1 × AI	−0.002 (0.007)	0.006 (0.010)	5.09 (6.02)	0.47 (1.30)
DESIGN 2: SESSION 2 × AI	0.003 (0.008)	−0.005 (0.009)	−1.67 (5.56)	−0.92 (1.26)
DESIGN 2: SESSION 3 × AI	−0.002 (0.008)	0.001 (0.009)	−1.08 (4.81)	−1.04 (1.29)
DESIGN 2: SESSION 4 × AI	0.006 (0.007)	0.004 (0.011)	−1.97 (4.86)	−0.28 (1.19)
DESIGN 3 × AI	−0.009 (0.009)	−0.001 (0.008)	- -	- -
CONTROL MEAN × AI	−0.036 (0.004)	0.004 (0.005)	6.28 (3.47)	1.69 (0.74)
PATHOLOGY FE	Yes	Yes	-	-
OBSERVATIONS	36280	36280	14635	14635
F-STAT	.63	.26	.46	.42
P>F-STAT	.67	.93	.76	.79

Note: Main specifications similar to table 2 with additional control variables for sequence number of a particular case and the experiment session. Due to the high volume of sequence numbers, we do not show them in this table but account for them.

Design 1 session dummy is omitted due to collinearity and is thus the control mean.